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Delayed creative destruction: How uncertainty shapes corporate assets

Contents

1 Big picture

- **Question.** How does uncertainty change firms' real asset decisions when assets are costly to adjust and costly to reverse? The paper asks not only whether uncertainty lowers investment, but also whether it slows the replacement of old capital with new, productive capital.
- **Setting.** The empirical setting is the global commercial shipping industry from 2006–2019. Shipping is ideal because ships are discrete capital units, their transactions are observable, and firms can adjust assets through four clearly measured margins: ordering new ships, buying used ships, selling used ships, and demolishing ships.
- **Main mechanism.** Higher uncertainty raises the option value of waiting. When a decision is hard to reverse, the firm is more likely to pause before committing. The paper applies this logic symmetrically to investment and disinvestment.
- **Why this goes beyond standard investment papers.** Many studies examine whether uncertainty reduces investment. This paper decomposes capital adjustment into investment and disinvestment, and then into margins with different irreversibility. This decomposition is the source of the paper's main contribution.
- **Core empirical result.** Higher uncertainty reduces both investment and disinvestment. The reductions are concentrated in the most irreversible margins: new ship orders on the investment side and ship demolitions on the disinvestment side.
- **Liquidity result.** Secondary-market liquidity moderates the effect. When resale markets are liquid, firms can more easily undo prior decisions, so uncertainty has a weaker effect. When resale markets are illiquid, uncertainty causes stronger delays.
- **Creative-destruction result.** The strongest distortions are in precisely the margins that renew the capital stock. Firms reduce investment in high-productivity and younger ships, and they delay disposing of low-productivity ships.
- **Firm-level consequence.** The asset-level delays accumulate into a firm-level portfolio effect. Under uncertainty, firms concentrate into fewer subsectors and hold older, smaller, less efficient fleets.
- **Identification reinforcement.** The paper supplements baseline panel regressions with a quasi-natural experiment: the 2009–2011 escalation in Somali piracy. Piracy increased uncertainty for Suez-transiting ship classes but left otherwise comparable ship classes less exposed.
- **Main takeaway.** Uncertainty slows creative destruction. It does not simply reduce the quantity of capital adjustment; it worsens the composition and productivity of corporate assets by delaying adoption of new capital and disposal of obsolete capital.

2 Data and measurement

2.1 Clarksons ship-level data: near-universe of global shipping firms

The paper uses ship-level data from Clarksons Research for 2006–2019. Clarksons is a leading maritime data provider, and the data cover the near-universe of commercial vessels in the global market. The database covers both public and private firms, and firms domiciled in more than one hundred countries.

- The ownership data identify each ship, its owner, owner country, builder, construction date, technical characteristics, and operating subsector.
- The transaction data identify new ship orders, used ship purchases and sales, resale prices, demolition events, and scrap values.
- The ownership history links ships to firms over time, allowing construction of firm-quarter and firm-subsector-quarter panels.

- The sample includes 3,966 unique firms. On average, these firms own seven ships in a quarter, worth about \$412 million in average purchase-price terms. Aggregate annual ship holdings are about \$378 billion.

Interpretation: The data are unusually well suited for testing real-options predictions because capital units are observable one by one. The researcher sees whether a firm orders a new ship, buys a used ship, sells a ship, demolishes a ship, and how the firm's fleet composition changes.

2.2 Sectors and subsectors: why the unit is firm-subsector-quarter

The paper divides ships into two broad sectors and eight size-based subsectors. Dry bulkers carry dry cargo such as iron ore, coal, minerals, and grain. Tankers carry crude oil and refined oil products.

- **Dry bulkers:** Handysize (DWT $<$ 40,000), Handymax (DWT 40,000–59,999), Panamax (DWT 60,000–109,999), and Capesize (DWT $\geq$ 110,000).
- **Tankers:** Medium Range/MR (DWT $<$ 45,000), Long Range 1/LR1 (DWT 45,000–79,999), Long Range 2/LR2 (DWT 80,000–159,999), and Very Large Crude Carriers/VLCC (DWT $\geq$ 160,000).
- These subsectors are economically segmented. Ship size determines feasible routes, cargo type, canal access, and secondary-market comparability.
- For example, Panamax bulkers are sized to traverse the Panama Canal, while Capesize ships are too large for the Suez and Panama Canals and must travel around the Cape of Good Hope.

Interpretation: The firm-subsector-quarter is the right unit because uncertainty, demand, and liquidity vary meaningfully by subsector. A firm may face high uncertainty in one ship market and low uncertainty in another, so firm-level aggregation alone would hide the relevant variation.

2.3 Investment and disinvestment outcomes

The paper measures capital allocation as rates scaled by the lagged stock of ships. It constructs both count-based and DWT-based measures, but the baseline uses the number of ships.

- **Investment rate:**

$$Investment_{i,j,t} = \frac{\#\text{ ships acquired by firm } i \text{ in subsector } j \text{ in quarter } t}{\#\text{ ships held by } i, j \text{ at } t-1}.$$

- Investment is decomposed into **new ship orders** and **used ship purchases**.
- **Disinvestment rate:**

$$Disinvestment_{i,j,t} = \frac{\#\text{ ships disinvested by firm } i \text{ in subsector } j \text{ in quarter } t}{\#\text{ ships held by } i, j \text{ at } t-1}.$$

- Disinvestment is decomposed into **demolition** and **used ship sales**.

Interpretation: The decomposition maps directly to reversibility. Used purchases and sales are secondary-market trades that can be partially reversed through another trade. New ship orders involve time-to-build, customization, and financing frictions. Demolition is final and cannot be reversed. Therefore, the model predicts that uncertainty should affect new orders and demolitions more strongly.

2.4 Uncertainty: option-implied volatility by shipping subsector

The baseline uncertainty variable is subsector-quarter uncertainty constructed from OptionMetrics.

1. For each shipping firm with publicly traded options, the authors identify, each day, the nearest-to-money American call option expiring at the end of the next quarter.
2. They obtain the annualized implied volatility of that option.
3. Daily implied volatility is averaged over the quarter for each firm.
4. Each firm is linked to the subsectors in which it operates during that quarter.
5. For each subsector-quarter, the firm-level implied volatilities are averaged across firms using the number of ships the firm operates in that subsector as weights.

$$\text{Uncertainty}_{j,t} = \sum_{f \in j,t} \omega_{f,j,t} \, IV_{f,t}, \qquad \omega_{f,j,t} = \frac{\#Ships_{f,j,t}}{\sum_{g \in j,t} \#Ships_{g,j,t}}.$$

Interpretation: This measure is forward-looking and specific to the relevant asset market. The paper also validates it with realized volatility of forward freight agreement prices; the correlation between ex ante option-implied uncertainty and ex post freight-rate volatility is about 0.82.

2.5 Demand and first-moment controls: ΔSubsectoral BDI and cash flow

Uncertainty is a second-moment concept, so the paper must avoid confusing it with changes in expected demand. It controls for first-moment conditions using forward freight agreement prices.

- ΔSubsectoral BDI is the quarterly percentage change in sectoral indices derived from forward freight agreement prices for routes commonly served by each subsector.
- These indices are quoted on the Baltic Exchange and obtained through Bloomberg.
- Cash Flow is income before extraordinary items plus depreciation and amortization divided by lagged total assets, averaged across firms in a subsector and weighted by the number of ships.

Interpretation: ΔSubsectoral BDI proxies for expected freight demand and expected investment opportunities. Cash Flow controls for internal financing conditions. Together, they help isolate uncertainty from demand and liquidity shocks.

2.6 Secondary-market liquidity and reversibility

The paper uses two baseline liquidity measures.

- **Price Premium:** the quarterly average resale price minus purchase price, divided by purchase price, across used ship transactions in the subsector. A higher premium implies a seller loses less when reselling and the market is more liquid.
- **Price Dispersion:** the reverse of the mean absolute deviation of resale prices divided by mean resale price. Lower dispersion means a more predictable resale price and therefore a more liquid market; the paper scales it so higher values represent higher liquidity.
- Additional market-thickness measures include number of ships, number of firms, number of transactions, and buyer-to-available-ship ratios.

Interpretation: Liquidity is the empirical proxy for reversibility. If the secondary market is thick, the firm can undo a decision by selling or buying in that market. If the market is thin, adjustment becomes more irreversible and uncertainty should matter more.

2.7 Productivity and vintage measures

The paper measures ship productivity with several observable technical characteristics.

- **Age:** newer ships embody newer technology, automation, safety, docking, fuel, and navigation improvements.
- **DWT:** dead-weight tonnage measures ship carrying capacity; larger ships can transport more cargo per trip.
- **RPM:** engine revolutions per minute; lower RPM indicates greater fuel efficiency.
- **Ratios:** Speed/RPM, DWT/RPM, and DWT/Speed summarize technical efficiency and fleet productivity.

For investment tests, a high-productivity ship is defined by DWT and RPM: high DWT and low RPM indicate higher productivity. For disinvestment tests, a high-productivity ship satisfies at least two of three criteria within a subsector-quarter: above-median DWT, below-median RPM, and below-median Age. Low-productivity ships are the complement.

Interpretation: These proxies let the paper connect capital allocation to creative destruction. New and efficient ships represent embodied technological progress; old and inefficient ships represent obsolete capital.

2.8 Firm-level composition measures

The firm-level tests aggregate across the subsectors in which a firm operates.

- Fleet concentration is measured by the log number of distinct subsectors and by HHI of ship holdings across subsectors.
- Fleet productivity is measured using average log Age, log DWT, log RPM, Speed/RPM, DWT/RPM, and DWT/Speed.
- Firm-level investment and disinvestment aggregate new, used, sale, and demolition decisions across all subsectors.

Interpretation: The firm-level tests ask whether asset-level responses offset or cumulate. If a firm merely shifts investment from one subsector to another, the firm-level fleet might not deteriorate. The paper finds that responses cumulate into narrower and less productive fleets.

3 Conceptual framework and empirical specifications

3.1 Real-options framework: uncertainty as a mean-preserving spread

The appendix formalizes uncertainty as a mean-preserving spread. This means a zero-mean, non-degenerate shock is added to the outcome distribution. The expected payoff may be unchanged, but the distribution becomes riskier.

Interpretation: This is the cleanest way to separate uncertainty from expected demand. The empirical analog is to control for first moments while estimating the effect of option-implied volatility.

3.2 Prediction 1: uncertainty delays investment and disinvestment

The firm holds a portfolio of ships and can adjust by buying, ordering, selling, or demolishing. When uncertainty rises and adjustment is costly to reverse, the value of waiting rises.

- For investment, waiting preserves the option to avoid buying an asset that may become unprofitable.
- For disinvestment, waiting preserves the option to keep an asset if market conditions improve.
- The same real-options logic applies to both sides of capital allocation.

Economic message: Uncertainty does not only suppress expansion. It also suppresses contraction and replacement. This is why the paper focuses on allocation and creative destruction rather than only investment.

3.3 Prediction 2: new orders and demolitions respond more than used trades

New orders and demolitions are more irreversible than used purchases and used sales.

- New orders require planning, customization, time to build, and financing commitments.
- Used purchases can be reversed by reselling the ship.
- Demolition is final: once a ship is scrapped, the firm cannot reacquire the same asset.
- Used sales can be partly reversed by buying similar ships in the secondary market.

Economic message: The prediction is not just "less investment." It is a ranking of margins: uncertainty should hit new orders and demolitions harder than secondary-market purchases and sales.

3.4 Prediction 3: liquidity mitigates uncertainty effects

If secondary markets are liquid, decisions are less costly to reverse. The paper therefore predicts that the negative effect of uncertainty is weaker when asset liquidity is high.

$$\frac{\partial^2 Y}{\partial \text{Uncertainty}\, \partial \text{Liquidity}} > 0$$

for investment and disinvestment outcomes where uncertainty has a negative main effect.

Interpretation: The interaction term is central. If $\beta_1 < 0$ and $\beta_3 > 0$, uncertainty delays adjustment, but the delay is smaller in liquid markets.

3.5 Prediction 4: uncertainty slows creative destruction

The model predicts that firms will reduce investment in more productive ships and disinvestment of less productive ships more strongly when uncertainty increases.

- Less investment in high-productivity ships delays adoption of new technology.
- Less demolition or sale of low-productivity ships delays disposal of old capital.
- The result is a decline in the productivity of the retained fleet.

Interpretation: This is the main conceptual contribution: uncertainty affects not only the quantity of capital adjustment, but also the quality and vintage of the capital stock.

3.6 Baseline empirical specification: uncertainty, liquidity, and asset allocation

The main specification is:

$$Y_{i,j,t} = \beta_1 \text{Uncertainty}_{j,t} + \beta_2 \text{Liquidity}_{j,t} + \beta_3 \text{Uncertainty}_{j,t} \times \text{Liquidity}_{j,t} + \theta' Controls_{i,j,t} + FE + \varepsilon_{i,j,t}. \quad (1)$$

- $Y_{i,j,t}$ is investment, disinvestment, component investment, component disinvestment, productivity-conditioned investment/disinvestment, or firm-level asset composition.
- $\text{Uncertainty}_{j,t}$ is option-implied uncertainty in subsector j in quarter t.
- $\text{Liquidity}_{j,t}$ is secondary-market liquidity in subsector j in quarter t.
- Controls include ΔSubsectoral BDI, Cash Flow, lagged average Age, lagged DWT, and lagged RPM.
- Fixed effects vary by specification, but the baseline includes country-by-year-quarter and firm-by-subsector fixed effects.
- Standard errors are double-clustered at the firm-subsector and year-quarter levels.

Interpretation: The fixed effects absorb persistent differences across firms and subsectors, while country-time fixed effects absorb macro shocks by headquarters country. The coefficients are identified from changes in uncertainty and liquidity within a firm-subsector over time, after controlling for demand and fleet characteristics.

3.7 Quasi-natural experiment: Somali piracy shock

The design-based specification is:

$$Y_{i,j,t} = \beta_1 Treat_j \times Piracy_t + \theta' Controls_{i,j,t} + Country \times YearQuarter + Firm \times Subsector + \varepsilon_{i,j,t}. \quad (2)$$

- $Treat_j = 1$ for Panamax bulkers and LR2 tankers.
- Control subsectors are Capesize bulkers and VLCC tankers.
- $Piracy_t = 1$ for 2009–2011, the period of escalated Somali pirate attacks.
- The treatment logic is based on physical route exposure: treated ship classes could transit the Suez Canal and were more exposed to Somali waters, while control ships were too large and usually avoided those routes.
- Matching pairs treated firm-subsectors with control firm-subsectors based on lagged fleet characteristics.

Interpretation: Somali piracy is useful because it created a sharp, externally driven increase in uncertainty for some ship classes but not others. The event-study version tests whether treated and control subsectors moved similarly before the shock.

3.8 Instrumental variables version

The IV version uses $Treat_j \times Piracy_t$ as an instrument for subsector uncertainty:

$$\text{Uncertainty}_{j,t} = \pi_0 + \pi_1 Treat_j \times Piracy_t + \pi_2' Controls_{i,j,t} + FE + u_{i,j,t}, \quad (3)$$

$$Y_{i,j,t} = \beta_1 \widehat{\text{Uncertainty}}_{j,t} + \theta' Controls_{i,j,t} + FE + \varepsilon_{i,j,t}. \quad (4)$$

Interpretation: The IV estimates are local to uncertainty induced by piracy. They help address concerns that baseline uncertainty could be correlated with omitted subsector-level demand or financing conditions.

4 Identification strategy

4.1 Baseline identification: within firm-subsector variation

The baseline regressions compare the same firm-subsector in periods of higher versus lower subsector uncertainty. Firm-by-subsector fixed effects absorb persistent differences in a firm's business model, fleet specialization, and subsector exposure.

- If a firm is permanently conservative in a certain ship type, that level difference is absorbed.
- If a subsector is permanently more volatile or less liquid, the fixed effect removes its average level.
- Identification comes from time variation in uncertainty and liquidity within firm-subsector cells.

Interpretation: This design is strong for removing time-invariant heterogeneity, but it still relies on controls and fixed effects to separate uncertainty from time-varying investment opportunities.

4.2 First-moment controls: separating uncertainty from expected demand

A major threat is that high uncertainty may coincide with weak expected demand. The paper addresses this by controlling for ΔSubsectoral BDI and Cash Flow.

- ΔSubsectoral BDI comes from forward freight agreement prices and captures expected route-level shipping rates.
- Cash Flow controls for the internal financing position of firms active in a subsector.
- Lagged fleet controls absorb changes in asset composition that may mechanically affect investment and disinvestment decisions.

Interpretation: If uncertainty coefficients remain after controlling for expected demand and cash flow, the estimates are less likely to reflect only deteriorating first moments.

4.3 Liquidity interaction as a mechanism test

The liquidity interaction is not merely a control. It is a direct test of the real-options mechanism.

- Real-options theory predicts stronger wait-and-see behavior when reversibility is low.
- Secondary-market liquidity measures reversibility.
- Therefore, uncertainty should matter more in illiquid secondary markets and less in liquid markets.

Interpretation: The positive uncertainty-liquidity interaction is the core mechanism evidence. It shows that uncertainty's effect is linked to the cost of reversing decisions, not just to general bad times.

4.4 Margin ranking: irreversibility across new orders, used purchases, demolition, and sales

The paper's strongest internal identification test is the comparison across margins.

- New orders are more irreversible than used purchases.
- Demolition is more irreversible than selling.
- The same uncertainty shock should have larger effects on the more irreversible margin if the real-options channel is correct.

Interpretation: This within-outcome ranking is difficult for many alternative explanations to match. A generic demand shock would reduce all margins, not necessarily concentrate in new orders and demolitions.

4.5 Somali piracy quasi-experiment

The piracy design strengthens causal interpretation through a shock that is external to firms' capital policies.

- Somali pirate attacks rose sharply in 2009–2011.
- Treated ship classes were those with Suez exposure: Panamax bulkers and LR2 tankers.
- Control ship classes were Capesize bulkers and VLCC tankers, which are too large for the Suez route and generally use different routes.
- The treatment is based on physical size and route feasibility, not on firms' endogenous expectations.
- Event studies check that treatment and control dynamics were not diverging before the piracy wave.

Interpretation: The identifying assumption is that, absent piracy, treated and control firm-subsectors would have followed parallel trends in investment and disinvestment after matching and controls.

4.6 Instrument strength and local interpretation

The first stage is strong: the piracy interaction sharply increases uncertainty in treated subsectors. The paper reports a first-stage F-statistic above conventional weak-instrument thresholds.

Interpretation: The IV estimates identify the effect of uncertainty for the particular uncertainty shock created by piracy. This local interpretation matters because piracy happened during a period of declining demand, so the authors are careful about external validity.

4.7 Robustness and alternative explanations

The paper checks several potential confounds.

- Alternative uncertainty measures: VIX, JLN macro uncertainty, EPU, GDP forecast dispersion, and realized volatility of freight-rate derivatives.
- Alternative liquidity and irreversibility measures: market thickness, buyer-to-ship ratios, transaction counts, and price dispersion.
- Exclusion of state-owned firms.
- Alternative clustering, including subsector clustering and wild bootstrap procedures.
- Firm-level uncertainty measures with subsector-by-year-quarter fixed effects.
- Financing constraints using size-age index and private-firm indicators.
- Lower-frequency estimation to address lumpy capital adjustment.
- DWT-based outcomes to ensure results are not only driven by small ships.

Interpretation: These tests show that the baseline pattern is not simply a result of broad macro uncertainty, state ownership, clustering choices, lumpy transactions, financing constraints, or measurement of uncertainty using only option-listed firms.

5 Tables and figures

5.1 Figure 1: Speed, RPM, and DWT of New Ships

Read:

- Panel (a) plots median speed and engine RPM for new ships ordered each year.
- Panel (b) plots median DWT for newly ordered ships.
- Over the sample, new ships become larger and more fuel efficient: DWT rises and RPM falls.
- Speed is relatively stable compared with RPM and DWT, which means the productivity improvement is not merely faster ships, but better carrying capacity and fuel efficiency.

Interpretation: The figure motivates why new orders are the key creative-destruction margin. New ships embody better technology. Delaying new orders means delaying adoption of larger, more efficient capital.

Economic message: The paper's productivity result requires that capital vintages differ. Figure 1 shows that they do: newer ships are technologically different from older ships.

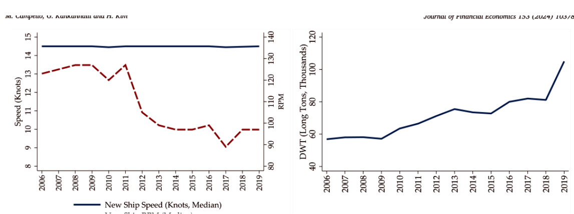

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Table 1
Descriptive Statistics. This table presents descriptive statistics for the main variables used in our empirical analyses over the 2006–2019 period. Dependent variables are indicated in normal-type font and independent variables are denoted in italics. The investment measures are calculated at the firm-subsector-quarter level by dividing the number (alternatively, dead-weight tons) of new, used, and the sum of new and used ships acquired by the lagged number (alternatively, dead-weight tons) of ships held, and are obtained from Clarksons Research. The disinvestment measures are calculated at the firm-subsector-quarter level by dividing the number (alternatively, dead-weight tons) of ships demolished, sold, and the sum of ships demolished and sold by the lagged number (alternatively, dead-weight tons) of ships held, and are obtained from Clarksons Research. *Age* is the average of the logarithm of the years since a ship was built, calculated across all ships in a given firm-subsector-quarter. *DWT* is the average of the logarithm of dead-weight tons of a ship, calculated across all ships in a given firm-subsector-quarter. *RPM* is the average of the logarithm of ship engine's revolutions per minute, calculated across all ships in a given firm-subsector-quarter. Δ*Subsectoral BDI* is the quarterly percentage change in the sectoral indices derived from forward freight agreement prices for routes commonly served by ships in each subsector quoted on the Baltic Exchange, and is obtained from Bloomberg. *Cash Flow* is the quarterly income before extraordinary items plus depreciation and amortization divided by lagged total assets, averaged across all firms operating in a given subsector weighted by the number of ships they operate in the subsector, and is obtained from Compustat and Orbis. *Uncertainty* is the quarterly average of the annualized daily implied volatility of nearest-to-money American call options expiring at the end of the next quarter, averaged across all firms operating in a given subsector, weighted by the number of ships they operate in the subsector, and is obtained from OptionMetrics. The *Liquidity* measures are defined as the price premium, which is the quarterly average resale price minus the purchase price divided by the purchase price, across all used ship transactions in a given subsector, and the (reverse of the) price dispersion, which is the quarterly mean absolute deviation of resale prices divided by the mean resale price of all transactions involving ships of a given subsector. All variables are winsorized at the 1% and 99% levels.

Variable	N	Mean	SD	Median	IQR
Investment Measures (Quarterly)					
Investment (Ships)/Lagged Number of Ships (%)	99,928	1.360	9.712	0	0
Investment (DWT)/Lagged DWT of Ships (%)	99,928	1.378	9.988	0	0
Investment (New Ships)/Lagged Number of Ships (%)	99,928	1.115	9.317	0	0
Investment (New DWT)/Lagged DWT of Ships (%)	99,928	1.128	8.614	0	0
Investment (Used Ships)/Lagged Number of Ships (%)	99,928	0.245	2.534	0	0
Investment (Used DWT)/Lagged DWT of Ships (%)	99,928	0.250	2.540	0	0
Disinvestment Measures (Quarterly)					
Disinvestment (Ships)/Lagged Number of Ships (%)	99,928	1.262	7.994	0	0
Disinvestment (DWT)/Lagged DWT of Ships (%)	99,928	1.257	8.004	0	0
Disinvestment (Ships Sold)/Lagged Number of Ships (%)	99,928	0.642	5.167	0	0
Disinvestment (DWT Sold)/Lagged DWT of Ships (%)	99,928	0.639	5.198	0	0
Disinvestment (Ships Demolished)/Lagged Number of Ships (%)	99,928	0.620	6.128	0	0
Disinvestment (DWT Demolished)/Lagged DWT of Ships (%)	99,928	0.618	5.218	0	0
Firm Fleet Characteristics (Quarterly)					
Age (Log # Years)	99,928	2.873	0.886	2.929	1.387
DWT (Log)	99,928	11.436	0.795	11.422	1.033
RPM (Log)	99,928	4.769	0.386	4.742	0.284

5.2 Table 1: Descriptive Statistics

Read:

- The main firm-subsector-quarter panel has 99,928 observations.
- Average quarterly investment is 1.360% of lagged ships; DWT-scaled investment is similar at 1.378%.
- Average quarterly disinvestment is 1.262% of lagged ships; DWT-scaled disinvestment is 1.283%.
- Median investment and disinvestment rates are zero, showing that capital adjustment is lumpy and discrete.
- The decomposition shows that new orders and used purchases are both observed, and demolitions and sales are both meaningful disinvestment margins.

Interpretation: Table 1 establishes why the empirical design needs a large panel. Most firm-subsector-quarters have no transaction, but the near-universe data provide enough events to estimate disaggregated margins.

Economic message: Lumpy investment is exactly where real-options logic matters. The table shows that shipping firms do not continuously adjust capital; they choose when to trigger discrete asset changes.

5.3 Table 2: Uncertainty, Asset Liquidity, and Investment and Disinvestment; Figure 2: Relationship between Uncertainty and Investment for Low and High Secondary Market Liquidity

Read:

- Panel A shows that uncertainty lowers total investment. With price-premium liquidity, the coefficient on uncertainty is −0.081 for total investment and −0.078 for new ships, both highly significant.
- Used purchases are much less affected: the used-investment coefficient is close to zero and insignificant in the price-premium specification.
- The uncertainty-liquidity interaction is positive and significant for total investment and new investment. High liquidity weakens the negative uncertainty effect.
- Panel B shows that uncertainty also lowers disinvestment. The coefficient on uncertainty is −0.208 for total disinvestment and −0.149 for demolition.
- Sales are less affected than demolition. This matches the irreversibility ranking.
- Figure 2 visualizes the investment result: at low liquidity, fitted new investment falls steeply as uncertainty rises; at high liquidity, the slope is flatter.

Interpretation: Table 2 is the main baseline evidence. It confirms the four-way structure of the theory: uncertainty delays adjustment, the effect is stronger on irreversible margins, liquidity weakens the delay, and investment/disinvestment both respond.

Identification role: The key identification value of Table 2 is the internal margin comparison. The same uncertainty measure is applied to new orders and used purchases, and to demolition and sales. The much stronger response on new orders and demolition is difficult to attribute to a generic decline in shipping demand.

Economic message: The table shows delayed creative destruction in its simplest form: firms do not just postpone expansion; they also postpone replacement and scrapping.

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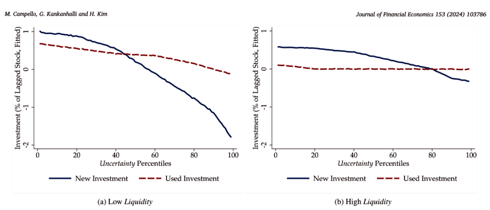
Table 2
Uncertainty, Asset Liquidity, and Investment and Disinvestment. This table reports output from Eq. (1). The dependent variable in Panel A is the investment rate, both the total rate and disaggregated into its components. The investment rate is calculated at the firm-subsector-quarter level by dividing the number of ships acquired (total, new ships acquired, or used ships acquired) by the lagged number of ships held. The dependent variable in Panel B is the disinvestment rate, both the total rate and disaggregated into its components. The disinvestment rate is calculated at the firm-subsector-quarter level by dividing the number of ships disinvested (total, ships demolished, or ships sold) by the lagged number of ships held. Δ*Subsectoral BDI* is the quarterly percentage change in the sectoral indices derived from forward freight agreement prices for routes commonly served by ships in each subsector quoted on the Baltic Exchange. *Cash Flow* is the quarterly income before extraordinary items plus depreciation and amortization divided by lagged total assets, averaged across all firms operating in a given subsector weighted by the number of ships they operate in the subsector. *Uncertainty* is the quarterly average of annualized daily implied volatility of nearest-to-money American call options expiring at the end of the next quarter, averaged across all firms operating in a given subsector, weighted by the number of ships they operate in the subsector. The *Liquidity* measures are defined as the price premium, which is the quarterly average resale price minus the purchase price divided by the purchase price, across all used ship transactions in a given subsector, and the (reverse of the) price dispersion, which is the quarterly mean absolute deviation of resale prices divided by the mean resale price of all transactions involving ships of a given subsector. Firm fleet controls are the average *Age*, *DWT*, and *RPM* of ships held by the firm-subsector in the lagged quarter. Fixed effects for headquarter country, firm, subsector, and year-quarter are included as indicated. All regressions are estimated over the 2006–2019 period. *t*-statistics are reported in parentheses, computed using robust standard errors clustered by firm-subsector and year-quarter.

	Panel A: Investment (Ships Acquired)					
	Total	New	Used	Total	New	Used
	(1)	(2)	(3)	(4)	(5)	(6)
Δ *Subsectoral BDI*	0.006***	0.006***	0.001	0.012***	0.009***	0.003
	(4.18)	(3.82)	(1.58)	(4.21)	(4.91)	(1.22)
Cash Flow	0.318	0.281	0.037	0.481	0.299	0.182
	(1.47)	(1.37)	(1.11)	(0.72)	(1.18)	(0.64)
Uncertainty	−0.081***	−0.078***	−0.004	−0.143***	−0.142***	−0.001
	(−3.23)	(−3.64)	(−0.17)	(−4.99)	(−5.17)	(−0.12)
Liquidity	0.221***	0.208***	0.013**	0.072***	0.049***	0.023***
	(3.77)	(3.85)	(2.11)	(5.88)	(6.76)	(7.92)
Uncertainty × *Liquidity*	0.021***	0.018***	0.002	0.014***	0.011***	0.003
	(3.56)	(3.84)	(0.20)	(4.24)	(5.16)	(1.15)
Liquidity Measure	Price Premium	Price Premium	Price Premium	Price Dispersion	Price Dispersion	Price Dispersion
Controls						
Firm-Subsector Fleet	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects						
Country × Year-Quarter	Yes	Yes	Yes	Yes	Yes	Yes
Firm × Subsector	Yes	Yes	Yes	Yes	Yes	Yes
Observations	99,928	99,928	99,928	99,928	99,928	99,928
Adjusted R^2	0.049	0.052	0.024	0.062	0.077	0.041

	Panel B: Disinvestment (Ships Disinvested)					
	Total	Demolished	Sold	Total	Demolished	Sold
	(1)	(2)	(3)	(4)	(5)	(6)
Δ *Subsectoral BDI*	−0.010*	−0.008**	−0.002*	−0.037*	−0.024*	−0.013*
	(−1.81)	(−2.10)	(−1.65)	(−1.91)	(−1.82)	(−1.80)
Cash Flow	−0.708	−0.687	−0.021	−0.897	−0.581	−0.316
	(−0.12)	(−0.78)	(−0.25)	(−0.65)	(−0.31)	(−0.96)
Uncertainty	−0.208***	−0.149***	−0.059*	−0.268***	−0.227***	−0.041
	(−3.99)	(−3.77)	(−1.73)	(−3.87)	(−4.31)	(−0.91)
Liquidity	0.142***	0.095**	0.047	0.124	0.129**	−0.005
	(2.82)	(2.46)	(1.46)	(1.40)	(2.00)	(−0.18)
Uncertainty × *Liquidity*	0.018***	0.013***	0.005*	0.014**	0.010**	0.004
	(4.05)	(3.75)	(1.83)	(2.41)	(2.28)	(0.49)
Liquidity Measure	Price Premium	Price Premium	Price Premium	Price Dispersion	Price Dispersion	Price Dispersion
Controls						
Firm-Subsector Fleet	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects						
Country × Year-Quarter	Yes	Yes	Yes	Yes	Yes	Yes
Firm × Subsector	Yes	Yes	Yes	Yes	Yes	Yes
Observations	99,928	99,928	99,928	99,928	99,928	99,928

(a) Table 2: Uncertainty, Asset Liquidity, and Investment and Disinvestment.



(b) Figure 2: Relationship between Uncertainty and Investment for Low and High Secondary Market Liquidity.

5.4 Figure 3: Frequency of Somali Attacks, Shipping Demand Levels, and Uncertainty; Table 3: Uncertainty, Investment, and Disinvestment: A Quasi-Natural Experiment; Figure 4: Investment and Disinvestment Dynamics around the Escalation in Somali Piracy

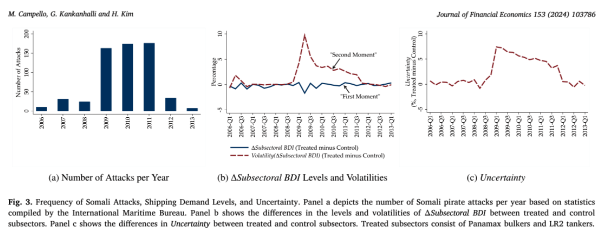
Read:

- Figure 3 documents the shock. Somali pirate attacks rise sharply in 2009–2011.
- Treated subsectors are Panamax bulkers and LR2 tankers; controls are Capesize bulkers and VLCC tankers.
- Figure 3 also shows that treated and control subsectors differ in uncertainty during the piracy period, while demand-level comparisons are included to evaluate first-moment concerns.
- Table 3 Panel A reports matched DiD estimates. $Treat \times Piracy$ reduces total investment by about 1.299 percentage points and new investment by about 1.197 percentage points.
- Panel A also shows reductions in disinvestment, with demolition again more affected than sales.
- Panel B instruments uncertainty with $Treat \times Piracy$. The second-stage coefficients on uncertainty are negative for total investment, new investment, total disinvestment, and demolition.
- Figure 4 shows event-study dynamics. Investment and disinvestment fall during the piracy escalation, with limited evidence of large pre-trends.

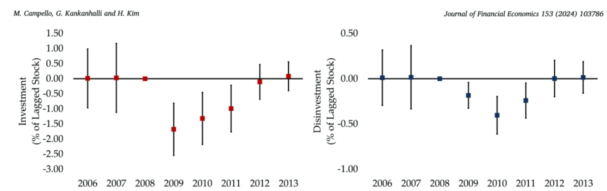
Interpretation: The piracy design provides causal reinforcement. The shock is external to individual firms' investment policies and is tied to route exposure created by ship size and canal feasibility.

Identification role: This section's identification strategy is: (1) define exposure using physical ship constraints, (2) match treated and control firm-subsectors on lagged fleet characteristics, (3) estimate DiD around 2009–2011, (4) inspect event-study pre-trends, and (5) use the same exposure as an instrument for uncertainty.

Economic message: The piracy evidence shows that an externally generated increase in uncertainty reduces capital adjustment in the same pattern as the baseline regressions.



(a) Figure 3: Frequency of Somali Attacks, Shipping Demand Levels, and Uncertainty.



(b) Figure 4: Investment and Disinvestment Dynamics around the Escalation in Somali Piracy.

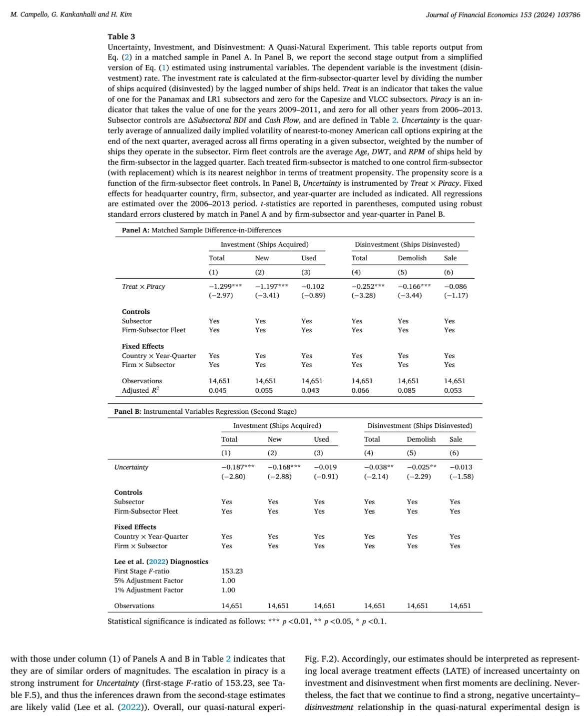


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Table 3

Uncertainty, Investment, and Disinvestment: A Quasi-Natural Experiment. This table reports output from Eq. (2) in a matched sample in Panel A. In Panel B, we report the second stage output from a simplified version of Eq. (1) estimated using instrumental variables. The dependent variable is the investment (disinvestment) rate. The investment rate is calculated at the firm-subsector-quarter level by dividing the number of ships acquired (disinvested) by the lagged number of ships held. *Treat* is an indicator that takes the value of one for the Panamax and LR1 subsectors and zero for the Capesize and VLCC subsectors. *Piracy* is an indicator that takes the value of one for the years 2009–2011, and zero for all other years from 2006–2013. Subsector controls are Δ*Subsectoral BDI* and *Cash Flow*, and are defined in Table 2. *Uncertainty* is the quarterly average of annualized daily implied volatility of nearest-to-money American call options expiring at the end of the next quarter, averaged across all firms operating in a given subsector, weighted by the number of ships they operate in the subsector. Firm fleet controls are the average *Age*, *DWT*, and *RPM* of ships held by the firm-subsector in the lagged quarter. Each treated firm-subsector is matched to one control firm-subsector (with replacement) which is its nearest neighbor in terms of treatment propensity. The propensity score is a function of the firm-subsector fleet controls. In Panel B, *Uncertainty* is instrumented by *Treat* × *Piracy*. Fixed effects for headquarter country, firm, subsector, and year-quarter are included as indicated. All regressions are estimated over the 2006–2013 period. t-statistics are reported in parentheses, computed using robust standard errors clustered by match in Panel A and by firm-subsector and year-quarter in Panel B.

Panel A: Matched Sample Difference-in-Differences

	Investment (Ships Acquired)			Disinvestment (Ships Disinvested)		
	Total	New	Used	Total	Demolish	Sale
	(1)	(2)	(3)	(4)	(5)	(6)
Treat × Piracy	−1.299***	−1.197***	−0.102	−0.252***	−0.166***	−0.086
	(−2.97)	(−3.41)	(−0.89)	(−3.28)	(−3.44)	(−1.17)
Controls						
Subsector	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Subsector Fleet	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects						
Country × Year-Quarter	Yes	Yes	Yes	Yes	Yes	Yes
Firm × Subsector	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,651	14,651	14,651	14,651	14,651	14,651
Adjusted R^2	0.045	0.055	0.043	0.066	0.085	0.053

Panel B: Instrumental Variables Regression (Second Stage)

	Investment (Ships Acquired)			Disinvestment (Ships Disinvested)		
	Total	New	Used	Total	Demolish	Sale
	(1)	(2)	(3)	(4)	(5)	(6)
Uncertainty	−0.187***	−0.168***	−0.019	−0.038**	−0.025**	−0.013
	(−2.80)	(−2.88)	(−0.91)	(−2.14)	(−2.29)	(−1.58)
Controls						
Subsector	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Subsector Fleet	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects						
Country × Year-Quarter	Yes	Yes	Yes	Yes	Yes	Yes
Firm × Subsector	Yes	Yes	Yes	Yes	Yes	Yes
Lee et al. (2022) Diagnostics						
First Stage F-ratio	153.23					
5% Adjustment Factor	1.00					
1% Adjustment Factor	1.00					
Observations	14,651	14,651	14,651	14,651	14,651	14,651

Statistical significance is indicated as follows: *** p <0.01, ** p <0.05, * p <0.1.

with those under column (1) of Panels A and B in Table 2 indicates that they are of similar orders of magnitudes. The escalation in piracy is a strong instrument for *Uncertainty* (first-stage F-ratio of 153.23, see Table F.5), and thus the inferences drawn from the second-stage estimates are likely valid (Lee et al. (2022)). Overall, our quasi-natural experi-

Fig. F.2). Accordingly, our estimates should be interpreted as representing local average treatment effects (LATE) of increased uncertainty on investment and disinvestment when first moments are declining. Nevertheless, the fact that we continue to find a strong, negative uncertainty–*disinvestment* relationship in the quasi-natural experimental design is

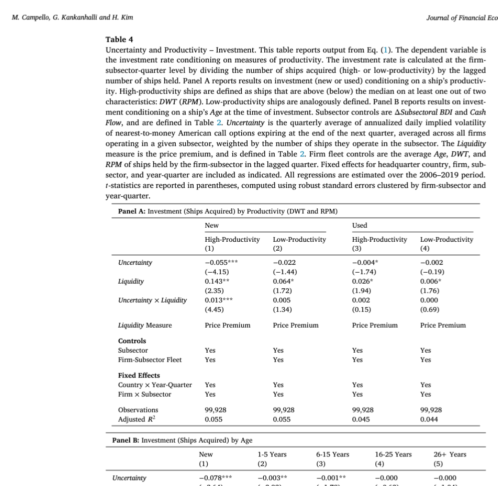

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Table 4

Uncertainty and Productivity – Investment. This table reports output from Eq. (1). The dependent variable is the investment rate conditioning on measures of productivity. The investment rate is calculated at the firm-subsector-quarter level by dividing the number of ships acquired (high- or low-productivity) by the lagged number of ships held. Panel A reports results on investment (new or used) conditioning on a ship's productivity. High-productivity ships are defined as ships that are above (below) the median on at least one out of two characteristics: *DWT* (*RPM*). Low-productivity ships are analogously defined. Panel B reports results on investment conditioning on a ship's *Age* at the time of investment. Subsector controls are Δ*Subsectoral BDI* and *Cash Flow*, and are defined in Table 2. *Uncertainty* is the quarterly average of annualized daily implied volatility of nearest-to-money American call options expiring at the end of the next quarter, averaged across all firms operating in a given subsector, weighted by the number of ships they operate in the subsector. The *Liquidity* measure is the price premium, and is defined in Table 2. Firm fleet controls are the average *Age*, *DWT*, and *RPM* of ships held by the firm-subsector in the lagged quarter. Fixed effects for headquarter country, firm, subsector, and year-quarter are included as indicated. All regressions are estimated over the 2006–2019 period. t-statistics are reported in parentheses, computed using robust standard errors clustered by firm-subsector and year-quarter.

Panel A: Investment (Ships Acquired) by Productivity (DWT and RPM)

	New		Used	
	High-Productivity (1)	Low-Productivity (2)	High-Productivity (3)	Low-Productivity (4)
Uncertainty	−0.055***	−0.022	−0.004*	−0.002
	(−4.15)	(−1.44)	(−1.74)	(−0.19)
Liquidity	0.143**	0.064*	0.026*	0.006*
	(2.35)	(1.72)	(1.94)	(1.76)
Uncertainty × Liquidity	0.013***	0.005	0.002	0.000
	(4.45)	(1.34)	(0.15)	(0.69)
Liquidity Measure	Price Premium	Price Premium	Price Premium	Price Premium
Controls				
Subsector	Yes	Yes	Yes	Yes
Firm-Subsector Fleet	Yes	Yes	Yes	Yes
Fixed Effects				
Country × Year-Quarter	Yes	Yes	Yes	Yes
Firm × Subsector	Yes	Yes	Yes	Yes
Observations	99,928	99,928	99,928	99,928
Adjusted R^2	0.055	0.055	0.045	0.044

Panel B: Investment (Ships Acquired) by Age

	New (1)	1-5 Years (2)	6-15 Years (3)	16-25 Years (4)	26+ Years (5)
Uncertainty	−0.078***	−0.003**	−0.001**	−0.000	−0.000

5.5 Table 4: Uncertainty and Productivity - Investment

Read:

- Panel A separates new and used investment by ship productivity measured using DWT and RPM.
- Uncertainty reduces new investment in high-productivity ships: the coefficient is −0.055 and statistically significant.

- New investment in low-productivity ships is less affected and statistically weaker.
- Used investment effects are small, consistent with used purchases being more reversible and less tied to adoption of new embodied technology.
- Panel B separates investment by age. The strongest negative effect is on new ships, then very young used ships; effects weaken as ship age increases.

Interpretation: Table 4 shows that uncertainty delays adoption of better capital. It is not merely that firms buy fewer ships; they buy fewer of the ships that would improve fleet productivity.

Economic message: This is the investment side of creative destruction. The technological frontier advances through new and young ships, and uncertainty slows movement toward that frontier.

5.6 Table 5: Uncertainty and Productivity - Disinvestment

Read:

- Table 5 separates disinvestment into high-productivity and low-productivity ships.
- Uncertainty has little effect on demolition and sale of high-productivity ships.
- Uncertainty strongly reduces demolition of low-productivity ships: the coefficient is -0.125 and highly significant.
- It also reduces sales of low-productivity ships, though the sale effect is smaller.
- Liquidity interactions are consistent with the baseline logic: more liquid markets mitigate uncertainty-driven delay.

Interpretation: Low-productivity ships are the assets that should be removed from the capital stock. The fact that uncertainty delays their demolition and sale means uncertainty slows the destruction side of creative destruction.

Economic message: The table complements Table 4. Firms delay both the creation of better capital and the destruction of obsolete capital.

5.7 Table 6: Uncertainty, Asset Liquidity, and Concentration of Firm Asset Holdings

Read:

- Panel A uses the log number of subsectors as the outcome. Uncertainty reduces the number of subsectors in which a firm operates.
- Panel B uses HHI of ship holdings. Uncertainty increases HHI, meaning firms concentrate their fleets.
- Liquidity interactions indicate that the concentration response is weaker when secondary markets are more liquid.

Interpretation: Table 6 moves from asset-level margins to portfolio structure. Under uncertainty, firms narrow their scope rather than expanding into or maintaining diversified subsector exposure.

Economic message: This matters because concentration can reduce firms' ability to redeploy assets across markets and can amplify productivity losses if the retained subsectors are not the most efficient uses of capital.

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Table 5
Uncertainty and Productivity – Disinvestment. This table reports output from Eq. (1). The dependent variable is the disinvestment rate conditioning on measures of productivity. The disinvestment rate is calculated at the firm-subsector-quarter level by dividing the number of ships disinvested (high- or low-productivity) by the lagged number of ships held. High-productivity ships are defined as ships that are above (below) the median on two out of three characteristics: *DWT* (*Age*, *RPM*). Low-productivity ships are analogously defined. Subsector controls are Δ*Subsectoral BDI* and *Cash Flow*, and are defined in Table 2. *Uncertainty* is the quarterly average of annualized daily implied volatility of nearest-to-money American call options expiring at the end of the next quarter, averaged across all firms operating in a given subsector, weighted by the number of ships they operate in the subsector. The *Liquidity* measure is the price premium, and is defined in Table 2. Firm fleet controls are the average *Age*, *DWT*, and *RPM* of ships held by the firm-subsector in the lagged quarter. Fixed effects for headquarter country, firm, subsector, and year-quarter are included as indicated. All regressions are estimated over the 2006–2019 period. *t*-statistics are reported in parentheses, computed using robust standard errors clustered by firm-subsector and year-quarter.

	High-Productivity Ships		Low-Productivity Ships	
	Demolition	Sale	Demolition	Sale
	(1)	(2)	(3)	(4)
Uncertainty	−0.024	−0.013	−0.125***	−0.046*
	(−0.46)	(−0.29)	(−4.89)	(−1.83)
Liquidity	0.018	0.021	0.077*	0.026
	(1.24)	(1.50)	(1.84)	(0.54)
Uncertainty × *Liquidity*	0.002	0.001	0.011***	0.004*
	(0.16)	(0.70)	(4.40)	(1.66)
Liquidity Measure	Price Premium	Price Premium	Price Premium	Price Premium
Controls				
Subsector	Yes	Yes	Yes	Yes

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Table 6
Uncertainty, Asset Liquidity, and Concentration of Firm Asset Holdings. This table reports output from a variant of Eq. (1) estimated at the firm-quarter level. The dependent variable in Panel A is the natural logarithm of the distinct number of subsectors a firm is exposed to, calculated at the firm-quarter level. The dependent variable in Panel B is the HHI of a firm's ship holdings across the subsectors it operates in, calculated at the firm-quarter level. Firm controls are *Cash Flow* and Δ*Subsectoral BDI*, and are defined in Table 2, except that they are averaged across all subsectors that a firm operates in weighted by the number of ships it operates in the subsector. *Uncertainty* is the quarterly average of annualized daily implied volatility of nearest-to-money American call options expiring at the end of the next quarter, averaged across all firms operating in a given subsector, weighted by the number of ships they operate in the subsector, averaged across all subsectors that a firm operates in weighted by the number of ships it operates in the subsector. The *Liquidity* measures are defined as the price premium, which is the quarterly average resale price minus the purchase price divided by the purchase price, across all used ship transactions in a given subsector, and the (reverse of the) price dispersion, which is the quarterly mean absolute deviation of resale prices divided by the mean resale price of all transactions involving ships of a given subsector. Both are averaged across all subsectors that a firm operates in weighted by the number of ships it operates in the subsector. Firm fleet controls are the average *Age*, *DWT*, and *RPM* of ships held by the firm in the lagged quarter. Fixed effects for headquarter country, firm, and year-quarter are included as indicated. All regressions are estimated over the 2006–2019 period. *t*-statistics are reported in parentheses, computed using robust standard errors clustered by firm and year-quarter.

Panel A: Log Number of Subsectors at the Firm Level

	(1)	(2)	(3)
Uncertainty	−0.006***	−0.005***	−0.002***
	(−7.39)	(−3.69)	(−3.60)
Liquidity		0.011	0.028
		(1.33)	(1.63)
Uncertainty × *Liquidity*		0.014**	0.008***
		(3.68)	(4.38)
Liquidity Measure	NA	Price Premium	Price Dispersion

7. Robustness and

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5.8 Table 7: Uncertainty, Asset Liquidity, and Productivity of Firm Asset Holdings

Read:

- Uncertainty increases average log Age of the fleet, implying firms hold older ships.
- Uncertainty lowers log DWT, implying fleets shift toward smaller ships.
- Uncertainty raises log RPM, implying less fuel-efficient engines.
- It lowers Speed/RPM, DWT/RPM, and DWT/Speed, all of which point toward lower fleet productivity.
- Liquidity interactions are generally in the direction of mitigation: liquid secondary markets dampen the adverse productivity effect.

Interpretation: Table 7 is the firm-level productivity confirmation. The asset-level delays observed in Tables 4 and 5 show up in the average productivity of the assets firms keep.

Economic message: The productivity loss is not a theoretical extrapolation. It is visible in the retained fleet: older, smaller, less efficient vessels.

5.9 Table 8: Uncertainty, Asset Liquidity, and Asset Allocation at the Firm Level

Read:

- Table 8 repeats the allocation analysis at the firm-quarter level.
- Panel A shows that firm-level investment falls with uncertainty, especially new investment.
- Panel B shows that firm-level disinvestment falls with uncertainty, especially demolition.
- Liquidity again mitigates the negative uncertainty effect.

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Table 7
Uncertainty, Asset Liquidity, and Productivity of Firm Asset Holdings. This table reports output from a variant of Eq. (1) estimated at the firm-quarter level. The dependent variables are the average logarithm of *Age*, logarithm of *DWT*, logarithm of *RPM*, *Speed/RPM*, *DWT/RPM*, and *DWT/Speed* ratios of ships held by a firm in a given quarter. Firm controls are *Cash Flow* and Δ*Subsectoral BDI*, and are defined in Table 2, except that they are averaged across all subsectors that a firm operates in weighted by the number of ships it operates in the subsector. *Uncertainty* is the quarterly average of annualized daily implied volatility of nearest-to-money American call options expiring at the end of the next quarter, averaged across all firms operating in a given subsector, weighted by the number of ships they operate in the subsector, averaged across all subsectors that a firm operates in weighted by the number of ships it operates in the subsector. The *Liquidity* measure is defined as the price premium, which is the quarterly average resale price minus the purchase price divided by the purchase price, across all used ship transactions in a given subsector. Both are averaged across all subsectors that a firm operates in weighted by the number of ships it operates in the subsector. Firm fleet controls are the average *Age*, *DWT*, and *RPM* of ships held by the firm in the lagged quarter. Note that in columns (1) through (3), we omit the lagged firm fleet control corresponding to the respective outcome variable. Fixed effects for headquarter country, firm, and year-quarter are included as indicated. All regressions are estimated over the 2006–2019 period. *t*-statistics are reported in parentheses, computed using robust standard errors clustered by firm and year-quarter.

Average Asset Characteristics at the Firm Level						
	Log Age	*Log DWT*	*Log RPM*	*Speed/RPM*	*DWT/RPM*	*DWT/Speed*
	(1)	(2)	(3)	(4)	(5)	(6)
Uncertainty	0.005***	−0.006***	0.002**	−0.538***	−0.025***	−0.462***
	(3.51)	(−2.81)	(2.15)	(−6.71)	(−3.37)	(−8.30)
Liquidity	−0.039***	0.004	−0.002	−2.426***	−0.227	−1.976***
	(−5.60)	(0.12)	(−0.51)	(−7.04)	(1.50)	(−8.49)
Uncertainty × *Liquidity*	−0.014***	0.018**	−0.009**	5.375***	0.248***	4.619***
	(−3.25)	(2.27)	(−2.22)	(6.71)	(3.37)	(8.30)
Liquidity Measure	Price Premium	Price Premium	Price Premium	Price Premium	Price Premium	Price Premium

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Table 8
Uncertainty, Asset Liquidity, and Asset Allocation at the Firm Level. This table reports output from a variant of Eq. (1) estimated at the firm-quarter level. The dependent variables are the investment and disinvestment rates. The investment rate is calculated at the firm-quarter level by dividing the number of ships acquired (new, used, and total) by the lagged number of ships held. The disinvestment rate is calculated at the firm-quarter level by dividing the number of ships disinvested (demolished, sold, and total) by the lagged number of ships held. Δ*Subsectoral BDI* is the quarterly percentage change in the sectoral indices derived from forward freight agreement prices for routes commonly served by ships in each subsector quoted on the Baltic Exchange, averaged across all subsectors that a firm operates in weighted by the number of ships it operates in the subsector. *Cash Flow* is the quarterly income before extraordinary items plus depreciation and amortization divided by lagged total assets, averaged across all firms operating in a given subsector weighted by the number of ships they operate in the subsector, averaged across all subsectors that a firm operates in weighted by the number of ships it operates in the subsector. *Uncertainty* is the quarterly average of annualized daily implied volatility of nearest-to-money American call options expiring at the end of the next quarter, averaged across all firms operating in a given subsector, weighted by the number of ships they operate in the subsector, averaged across all subsectors that a firm operates in weighted by the number of ships it operates in the subsector. The *Liquidity* measure is defined as the price premium, which is the quarterly average resale price minus the purchase price divided by the purchase price, across all used ship transactions in a given subsector. Both are averaged across all subsectors that a firm operates in weighted by the number of ships it operates in the subsector. Firm fleet controls are the average *Age*, *DWT*, and *RPM* of ships held by the firm in the lagged quarter. Fixed effects for headquarter country, firm, and year-quarter are included as indicated. All regressions are estimated over the 2006–2019 period. *t*-statistics are reported in parentheses, computed using robust standard errors clustered by firm and year-quarter.

Panel A: Investment (Ships Acquired) at the Firm Level						
	Total		New		Used	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ *Subsectoral BDI*	0.005***	0.006***	0.004***	0.003***	0.001***	0.003**
	(3.55)	(3.93)	(3.07)	(4.10)	(3.37)	(2.02)
Cash Flow	0.012*	0.004***	0.009**	0.002***	0.003**	0.001**
	(1.66)	(4.75)	(2.38)	(4.74)	(2.24)	(2.40)
Uncertainty	−0.067**	−0.119**	−0.051***	−0.099***	−0.016	−0.020
	(−2.64)	(−2.39)	(−3.12)	(−3.48)	(−1.08)	(−0.66)
Liquidity		2.607***		2.490***		0.117
		(3.47)		(3.67)		(1.34)
Uncertainty × *Liquidity*		0.202**		0.184**		0.018
		(2.33)		(2.53)		(1.44)
Liquidity Measure	NA	Price Premium	NA	Price Premium	NA	Price Premium

Interpretation: The firm-level results show that the baseline firm-subsector patterns do not merely reflect shifting activity from one subsector to another. The entire firm adjusts less.

Economic message: Uncertainty changes the whole corporate asset portfolio, not only marginal decisions in individual ship markets.

5.10 Table 9: Aggregate versus Sectoral Uncertainty and Investment and Disinvestment; Figure A.1: Asset Market Thickness and Liquidity

Read:

- Table 9 compares broad uncertainty measures with sector-specific uncertainty measures.
- VIX, JLN macro uncertainty, EPU, and GDP forecast dispersion do not reproduce the full baseline pattern as strongly.
- Realized volatility of subsectoral freight-rate changes does reproduce the key uncertainty and interaction effects.
- Figure A.1 shows that thicker secondary markets are associated with higher price premiums and lower price dispersion, supporting the interpretation of liquidity as market thickness.

Interpretation: The robustness evidence indicates that the relevant uncertainty is the uncertainty faced in the shipping asset market, not generic macro uncertainty. It also validates the liquidity proxies as measures of reversibility.

Economic message: Asset-market-specific frictions are central to the paper. The macro effect of uncertainty operates through the microstructure of capital markets for ships.

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Table 9
Aggregate *versus* Sectoral Uncertainty and Investment and Disinvestment. This table reports output from Eq. (1). The dependent variables are the investment and disinvestment rates. The investment (disinvestment) rate is calculated at the firm-subsector-quarter level by dividing the number of ships acquired (disinvested) by the lagged number of ships held. *VIX* is the quarterly Volatility Index from the CBOE. *JLN Uncertainty* is the quarterly uncertainty index from Jurado et al. (2015). *EPU* is the quarterly US Economic Policy Uncertainty Index from Baker et al. (2016). *GDP Dispersion* is the quarterly GDP forecast dispersion from the Survey of Professional Forecasters compiled by the Federal Reserve Bank of Philadelphia. *Volatility*(Δ*Subsectoral BDI*) is the quarterly realized volatility of Δ*Subsectoral BDI*. Subsector controls consist of Δ*Subsectoral BDI* and *Cash Flow*. See Table 2 for definitions of these variables. The *Liquidity* measure is defined as the price premium, which is the average resale price minus the purchase price divided by the purchase price, across all used ship transactions in a given subsector. Firm fleet controls are the average *Age*, *DWT*, and *RPM* of ships held by the firm-subsector in the lagged quarter. Fixed effects for firm and subsector are included as indicated. All regressions are estimated over the 2006–2019 period. *t*-statistics are reported in parentheses, computed using robust standard errors clustered by firm-subsector and year-quarter.

	Investment	Disinvestment
	(1)	(2)
Liquidity	0.366***	−0.044
	(3.40)	(−0.45)
VIX	−0.057	−0.018
	(−0.85)	(−0.28)
VIX × *Liquidity*	0.018	0.002
	(0.85)	(1.45)
JLN Uncertainty	−0.184	−0.125
	(−0.63)	(−1.22)
JLN Uncertainty × *Liquidity*	0.355	0.058
	(0.63)	(0.50)
EPU	−0.005	−0.001
	(−1.22)	(−0.41)
EPU × *Liquidity*	0.004	0.002
	(0.95)	(0.91)
GDP Dispersion	−1.198	−1.240
	(−0.97)	(−1.01)
GDP Dispersion × *Liquidity*	4.877	6.585
	(0.42)	(0.71)
Volatility(Δ*Subsectoral BDI*)	−0.163***	−0.039***
	(−8.32)	(−3.51)
Volatility(Δ*Subsectoral BDI*) × *Liquidity*	0.014***	0.021***
	(3.15)	(3.90)
Liquidity Measure	Price Premium	Price Premium
Controls		
Subsector	Yes	Yes
Firm-Subsector Fleet	Yes	Yes
Fixed Effects		
Firm × Subsector	Yes	Yes
Observations	99,928	99,928

(a) Table 9: Aggregate versus Sectoral Uncertainty and Investment and Disinvestment.

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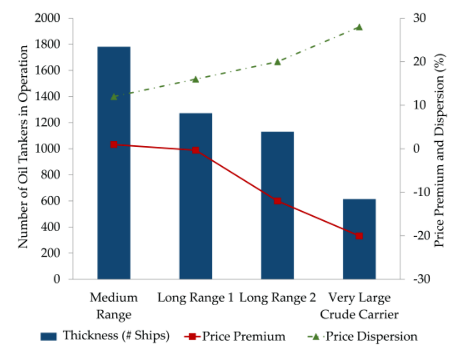

Fig. A.1. Asset Market Thickness and Liquidity. This figure shows the thickness of the secondary ship market, measured by the average number of ships in each subsector of Oil Tankers (blue vertical bars, primary Y-axis), and the liquidity, measured by the average percentage resale price premium (red connected squares, secondary Y-axis) and the resale price dispersion (green connected triangles, secondary Y-axis). Resale price dispersion is defined as the mean absolute deviation of resale prices divided by the mean resale price.

(b) Figure A.1: Asset Market Thickness and Liquidity.

6 Interpretive synthesis

6.1 What is the mechanism?

The mechanism is a real-options mechanism operating through reversibility. When uncertainty rises, firms value waiting. Waiting is especially valuable when decisions are hard to reverse. Therefore uncertainty lowers both investment and disinvestment, with the strongest effects on new orders and demolitions.

6.2 Why is this creative destruction?

Creative destruction requires two simultaneous movements: adopting new productive capital and discarding old unproductive capital. The paper shows that uncertainty slows both. Firms buy fewer productive and young ships, and they also remove fewer low-productivity ships. The retained capital stock becomes older and less efficient.

6.3 Why does liquidity matter?

Liquidity reduces the cost of making a mistake. If a firm can resell a ship easily, buying is less irreversible. If it can buy a similar ship later, selling is less irreversible. A thick secondary market therefore weakens the wait-and-see effect.

6.4 What is the role of the Somali piracy experiment?

The piracy episode helps show that the baseline result is not only a correlation between volatility and weak demand. Piracy created a route-specific uncertainty shock that affected ships capable of Suez transit more than comparable ship types. The matched DiD, event study, and IV results show the same pattern as the baseline: uncertainty reduces new investment and demolition.

7 Short summary

- The paper uses near-universal global shipping data from 2006–2019.
- It observes firm-level ownership and ship-level transactions: new orders, used purchases, sales, and demolitions.
- Uncertainty is measured using option-implied volatility of shipping firms and aggregated to subsector-quarters.
- Demand is controlled using forward freight agreement price changes, ΔSubsectoral BDI.
- Liquidity is measured using secondary-market resale price premiums and price dispersion.
- Uncertainty reduces investment and disinvestment.
- The effect is strongest for new ship orders and ship demolitions, the most irreversible margins.
- Secondary-market liquidity mitigates the negative effect.
- Somali piracy provides quasi-experimental evidence that route-specific uncertainty reduces capital allocation.
- Uncertainty reduces investment in high-productivity and young ships.
- Uncertainty delays demolition and sale of low-productivity ships.
- Firms exposed to uncertainty concentrate into fewer subsectors.
- Their fleets become older, smaller, and less fuel-efficient.
- Robustness tests show that sector-specific uncertainty matters more than broad macro uncertainty.
- The central conclusion is that uncertainty delays creative destruction.

8 Conclusion

8.1 Core conclusion

Uncertainty slows the process by which firms renew their capital stock. It reduces both acquisition and disposal of ships, with the strongest effects on new ship orders and demolitions. This means uncertainty delays both the adoption of new technology and the retirement of old technology.

8.2 Economic intuition

The key intuition is the option value of waiting. When future market conditions are uncertain, a firm may prefer to delay an irreversible decision. A new ship order commits the firm to a costly, customized investment. Demolition permanently removes an asset from the fleet. Secondary-market liquidity lowers the cost of reversal, so it weakens the option value of waiting.

8.3 Why the paper matters

The paper connects uncertainty to productivity through the composition of capital. Many uncertainty papers stop at investment quantities. This paper shows that uncertainty also changes which assets firms buy, which assets they scrap, and what their portfolios look like afterward.

8.4 Policy and managerial implication

Policies or market institutions that improve asset-market liquidity can reduce the real effects of uncertainty. If firms can more easily buy and sell capital assets, uncertainty is less likely to freeze capital reallocation. For managers, the evidence suggests that maintaining access to liquid asset markets can preserve flexibility without fully stopping renewal of productive assets.

8.5 Final takeaway

Uncertainty does not merely postpone spending. It postpones creative destruction. The cost is a less dynamic and less productive corporate asset base.